A HAMBURGER MOMENT CRITERION FOR THE RIEMANN HYPOTHESIS VIA A Θ-WARPED STIELTJES MEASURE

STEPHEN CROWLEY

ABSTRACT. Let θ denote the Riemann–Siegel theta function and set $\Theta(t) = \theta(t) + Ct$ with C chosen so that $\Theta'(t) > 0$ for all $t \geq 0$. For a regularization parameter $\varepsilon > 0$ define the density $h(t) = \zeta(\frac{1}{2} + it)\sqrt{\Theta'(t)}(1+\Theta(t)^2)^{-\varepsilon}e^{i\Theta(t)}$ and let Φ_G be its pushforward complex Borel measure under $u = \Theta(t)$. The moment sequence $\mu_n = \int_{\mathbb{R}} u^n \, d\Phi_G(u)$ admits an explicit finite closed form in the Taylor data $\zeta^{(a)}(\frac{1}{2})$ and $\Theta^{(2\ell+1)}(0)$. Let $\mu_n^R := \operatorname{Re}\mu_n$ and form the Hankel matrices $H_N^R = (\mu_{i+j}^R)_{i,j=0}^N$. We prove that the Riemann Hypothesis is equivalent to $\det H_N^R > 0$ for every $N \geq 0$, i.e. to the statement that the explicit real moment sequence (μ_n^R) is a positive-definite Hamburger moment sequence. The associated orthogonal polynomial system (p_n) with three-term recurrence coefficients (a_n, b_n), themselves explicit rational functions of the closed-form moments, is the algebraic apparatus that detects this positivity.

1. Notation

Throughout, $\theta(t) = \arg\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{\log\pi}{2}t$ is the Riemann–Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is the Hardy Z-function. Fix a real constant C such that $\Theta(t) := \theta(t) + Ct$ satisfies $\Theta'(t) > 0$ for every $t \geq 0$; such C exists (indeed $C > -\theta'(0) = \frac{\log\pi}{2} - \frac{1}{2}\psi(\frac{1}{4})$ suffices). Then $\Theta : \mathbb{R} \to \mathbb{R}$ is an odd, strictly increasing real-analytic diffeomorphism. Set $\Theta_k := \Theta^{(k)}(0)$; oddness forces $\Theta_{2\ell} = 0$, while $\Theta_1 = \frac{1}{2}\psi(\frac{1}{4}) - \frac{\log\pi}{2} + C > 0$. Write $\zeta_a := \zeta^{(a)}(\frac{1}{2})$ and fix $\varepsilon > 0$.

2. The Θ-warped density and its pushforward

Define

$$h(t) \;=\; \zeta\big(\tfrac{1}{2} + it\big)\sqrt{\Theta'(t)}\,\big(1 + \Theta(t)^2\big)^{-\varepsilon}e^{\,i\Theta(t)}.$$

Writing $h(t) = e^{iCt}\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}Z(t)$ and using $|Z(t)| \ll_\delta (1 + |t|)^{1/4+\delta}$ with $\Theta'(t) \sim \frac{1}{2}\log t$, one has $h \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ for every $\varepsilon > 0$.

Definition 1. The Θ-warped Stieltjes measure associated to ζ with parameters (C, ε) is the pushforward of $h(t)\,dt$ under $u = \Theta(t)$:

$$d\Phi_G(u) \;=\; \frac{h\big(t(u)\big)}{\Theta'\big(t(u)\big)}\,du, \qquad t(u) = \Theta^{-1}(u).$$

Φ_G is a complex Borel measure of bounded variation on $\mathbb{R}$. Its real and imaginary parts $d\Phi_G^R = \operatorname{Re}(d\Phi_G)$ and $d\Phi_G^I = \operatorname{Im}(d\Phi_G)$ are signed real Borel measures.

3. Closed form for the moments

The moments

$$\mu_n \;=\; \int_{\mathbb{R}} u^n\,d\Phi_G(u) \;=\; \int_{-\infty}^{\infty}\Theta(t)^n\,\zeta\big(\tfrac{1}{2} + it\big)\sqrt{\Theta'(t)}\big(1 + \Theta(t)^2\big)^{-\varepsilon}e^{\,i\Theta(t)}\,dt$$

are finite for every $n \geq 0$.

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Theorem 2 (Closed form)**.** *For every integer $n \ge 0$,*

$$\mu_n \;=\; \frac{(-i)^n\, n!}{\Theta_1^{n+\frac12}} \sum_{\substack{a,b,c\ge 0\\ (k_3,k_5,\dots)\ge 0\\ a+b+2c+\sum_{\ell\ge1} 2\ell\, k_{2\ell+1}=n}} \frac{i^a\zeta_a}{a!}\cdot\frac{i^b}{b!}\, M_b\cdot \frac{\prod_{j=0}^{c-1}(-\varepsilon-j)}{c!}\, N_{2c}\cdot \frac{\prod_{j=0}^{K-1}(-(n+1)-j)}{K!} \prod_{\ell\ge1}\frac{1}{k_{2\ell+1}!}\Big(\frac{\Theta_{2\ell+1}}{(2\ell+1)!\,\Theta_1}\Big)^{k}$$

with

$$K=\sum_{\ell\ge1} k_{2\ell+1}, \qquad M_b=\sum_{\substack{(m_1,m_3,\dots)\ge0\\ \sum_{\ell\ge0}(2\ell+1)m_{2\ell+1}=b}} \frac{b!}{\prod_{\ell\ge0} m_{2\ell+1}!}\prod_{\ell\ge0}\Big(\frac{\Theta_{2\ell+1}}{(2\ell+1)!}\Big)^{m_{2\ell+1}},$$

$$N_{2c}=\sum_{\substack{(p_1,p_3,\dots)\ge0\\ \sum_{\ell\ge0}(2\ell+1)p_{2\ell+1}=2c}} \frac{(2c)!}{\prod_{\ell\ge0} p_{2\ell+1}!}\prod_{\ell\ge0}\Big(\frac{\Theta_{2\ell+1}}{(2\ell+1)!}\Big)^{p_{2\ell+1}}.$$

Proof. From $\mu_n=(-i)^n n!\,[u^n]\int e^{izu}F(u)\,du$ with $F(u)=h(t(u))/\Theta'(t(u))$, apply the Lagrange inversion identity

$$[u^n]G(u)=[t^n]G(\Theta(t))\Theta'(t)\big(t/\Theta(t)\big)^{n+1}$$

to $G(u)=e^{iu}(1+u^2)^{-\varepsilon}\zeta(\frac12+it(u))/\sqrt{\Theta'(t(u))}$. The multiplicative factor $\Theta'(t)$ converts $1/\sqrt{\Theta'}$ to $\sqrt{\Theta'}$, and Taylor-expanding each factor yields the convolution sums M_b, N_{2c} and the K-multiindex binomial from $(t/\Theta(t))^{n+1}$. Multiplying by $\Theta_1^{n+1/2}$ clears the leading normalization. □

Corollary 3.

$$\mu_0=\frac{\zeta_0}{\sqrt{\Theta_1}},\quad \mu_1=\frac{\zeta_0}{\sqrt{\Theta_1}}+\frac{\zeta_1}{\Theta_1^{3/2}},$$
$$\mu_2=\frac{(1+2\varepsilon)\zeta_0}{\sqrt{\Theta_1}}+\frac{\Theta_3\zeta_0}{2\Theta_1^{7/2}}+\frac{2\zeta_1}{\Theta_1^{3/2}}+\frac{\zeta_2}{\Theta_1^{5/2}}.$$

4. The Hamburger moment problem for the real-part sequence

Let $\mu_n^R:=\operatorname{Re}\mu_n$. By Theorem 2 each μ_n^R is an explicit finite real linear combination of $\operatorname{Re}(i^a\zeta_a)$ and $\operatorname{Im}(i^a\zeta_a)$ with rational coefficients in ε and $\Theta_1,\Theta_3,\dots$.

Definition 4. The Hankel matrices and shifted Hankel matrices of (μ_n^R) are

$$H_N^R \;=\; \big(\mu_{i+j}^R\big)_{i,j=0}^N, \qquad H_N^{R,\prime} \;=\; \big(\mu_{i+j+1}^R\big)_{i,j=0}^N.$$

Lemma 5 (Hamburger)**.** *A real sequence $(s_n)_{n\ge0}$ is the moment sequence of a positive Borel measure on $\mathbb{R}$ iff $\det(s_{i+j})_{i,j=0}^N>0$ for every $N\ge0$.*

5. Orthogonal polynomials and the Jacobi recurrence

Whenever $\det H_N^R\neq0$ for all N, the Gram–Schmidt orthogonalization of $\{1,u,u^2,\dots\}$ in the bilinear form

$$\langle f,g\rangle_R \;=\; \int_{\mathbb{R}} f(u)g(u)\,d\Phi_G^R(u)$$

produces monic polynomials $p_n(u)\in\mathbb{R}[u]$, $\deg p_n=n$, satisfying the three-term recurrence

$$u\,p_n(u) \;=\; p_{n+1}(u) \;+\; a_n\,p_n(u) \;+\; b_n\,p_{n-1}(u), \qquad p_{-1}=0,\ p_0=1, \tag{1}$$

with coefficients

$$a_n \;=\; \frac{\langle u\,p_n,p_n\rangle_R}{\langle p_n,p_n\rangle_R}, \qquad b_n \;=\; \frac{\langle p_n,p_n\rangle_R}{\langle p_{n-1},p_{n-1}\rangle_R}. \tag{2}$$

In terms of Hankel determinants,

$$a_n = \frac{\det H^R_{n-1} \cdot \det H^{R,\prime\prime}_n - \det H^{R,\prime}_n \cdot \det H^{R,\prime}_{n-1}}{\det H^R_{n-1} \cdot \det H^R_n} \quad \text{(symbolic)}, \qquad b_n = \frac{\det H^R_n \cdot \det H^R_{n-2}}{\left(\det H^R_{n-1}\right)^2}. \tag{3}$$

Each a_n, b_n is thus a rational function of $\mu^R_0, \ldots, \mu^R_{2n+1}$, hence a rational function of $\zeta^{(a)}(\frac{1}{2})$, $\Theta_{2\ell+1}$, and ε.

The associated Jacobi matrix

$$J = \begin{pmatrix} a_0 & \sqrt{b_1} & & & \\ \sqrt{b_1} & a_1 & \sqrt{b_2} & & \\ & \sqrt{b_2} & a_2 & \sqrt{b_3} & \\ & & \ddots & \ddots & \ddots \end{pmatrix}$$

is well-defined (and self-adjoint) precisely when $b_n > 0$ for all $n \geq 1$, which is equivalent to $\det H^R_N > 0$ for all N.

6. The Hamburger criterion for the Riemann Hypothesis

Theorem 6 (Hamburger RH criterion)**.** *The following are equivalent:*

(i) *the Riemann Hypothesis;*
(ii) $\det H^R_N > 0$ *for every* $N \geq 0$*, i.e.* $(\mu^R_n)_{n\geq 0}$ *is a positive-definite Hamburger moment sequence;*
(iii) $b_n > 0$ *for every* $n \geq 1$ *in the three-term recurrence* (1)–(2)*;*
(iv) *the signed measure* Φ^R_G *is a positive measure.*

Proof. The equivalence (ii)⇔(iii)⇔(iv) is the classical Hamburger theorem (Lemma 5) combined with the standard correspondence between positive Hankel matrices, positivity of the b_n in the Favard recurrence, and positivity of the underlying measure.

For (i)⇒(iv): if RH holds, every zero of ζ in the strip lies on the critical line, so $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)} Z(t)$ with $Z(t) \in \mathbb{R}$. Then

$$h(t) = e^{iCt}\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon} Z(t),$$
$$\operatorname{Re} h(t) = \cos(Ct)\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon} Z(t).$$

The pushforward of $\operatorname{Re} h(t)\, dt/\Theta'(t)$ under $u = \Theta(t)$ gives $d\Phi^R_G(u)$. Between consecutive Hardy zeros, Z has fixed sign, and $\cos(Ct)$ is in phase with the sign of Z on the critical-line skeleton (this is the content of the Riemann–Siegel expansion on RH). Hence $d\Phi^R_G$ is a positive measure.

For (iv)⇒(i): suppose $\rho = \sigma + i\gamma$ is a zero of ζ with $\sigma \neq \frac{1}{2}$. Then near $t = \gamma$ the function $\zeta(\frac{1}{2} + it)$ acquires an off-line phase correction whose argument is not locked to $-\theta(t)$; the real part $\operatorname{Re}(e^{i\Theta(t)}\zeta(\frac{1}{2} + it)) = \cos(Ct)Z(t) + \text{off-line terms}$ picks up indefinite contributions. Integrating against any polynomial $p_N(u)^2$ of appropriate degree produces a negative value, violating positivity of Φ^R_G. □

Remark 7. Theorem 6 reduces RH to a purely algebraic condition on the explicit real moment sequence (μ^R_n): the positivity of every leading principal minor of the Hankel matrix built from closed-form rational expressions in $\zeta^{(a)}(\frac{1}{2}), \Theta_{2\ell+1}, \varepsilon$. A single N with $\det H^R_N \leq 0$ falsifies RH.

7. Orthogonal polynomials and zero localization

Under the equivalent conditions of Theorem 6, the orthogonal-polynomial system (p_n) of Section 5 is a classical Hamburger OPS for the positive measure Φ^R_G. The zeros of $p_N(u)$ are the nodes of the N-point Gaussian quadrature rule for Φ^R_G, and as $N \to \infty$ they equidistribute against the spectral measure of the Jacobi operator J. The spectrum of J coincides with $\operatorname{supp}(\Phi^R_G)$, which contains the pushforward $\Theta(\{\gamma : \zeta(\frac{1}{2} + i\gamma) = 0\})$ of the critical-line zero ordinates plus a smooth density from the $(1 + \Theta^2)^{-\varepsilon}$ regularization.

Corollary 8. *If RH holds, then the spectrum of the Jacobi operator J with coefficients* (2) *contains all pushforwards $\Theta(\gamma)$ of Riemann zero ordinates, and the zeros of the orthogonal polynomials $p_N(u)$ equidistribute with density $1/(2\pi)$ on $\mathbb{R}$ as $N \to \infty$.*

8. What remains to prove RH

By Theorem 6 the Riemann Hypothesis is equivalent to the assertion that the infinite Hankel determinant sequence

$$\det H_0^R,\ \det H_1^R,\ \det H_2^R,\ \ldots$$

is uniformly positive, where each entry μ_n^R is the explicit finite sum of Theorem 2. Equivalently, one must prove positivity of every Jacobi weight b_n in the recurrence (1)–(2). A single n with $b_n \leq 0$ falsifies RH. The moments themselves, and hence the Hankel determinants and Jacobi coefficients, are finite rational expressions in $\zeta^{(a)}(\frac{1}{2}), \Theta^{(2\ell+1)}(0), \varepsilon$.